

Effects of Graded Treatment with Phthalates, 4-Alkylphenols, and 17 α -ethynylestradiol (EE₂) on Feminization of Male Fish

Abstract

The purpose of this meta-analysis is to examine three xenoestrogens—phthalates, 4-alkylphenols, and 17 α -ethynylestradiol (EE₂) and determine which of the three is the most harmful and feminizing to fish. Xenoestrogens are of particular interest because they are a common environmental pollutant arising from a variety of sources. As xenoestrogens alter the reproductive systems of fish, this in turn can damage ecosystems as decreased fertility and sex-bias may cause a population collapse. We cross-examined experiments involving microgram-scale dosages of the aforementioned chemicals and their effects on two biomarkers--endogenous Vitellogenin (VTG) production and the Gonadosomatic Index (GSI) of varying species of fish. We assumed that all or most species of chordate fish have similar-functioning endocrine systems, and thus, are affected similarly by xenoestrogens. Results were compared using numerous analysis of variance tests (ANOVA) to determine if relationships and differences between the chemicals' effects and concentrations are significant. From this meta-analysis, we found that VTG production was increased with increasing concentrations of all three xenoestrogens, and no obvious trend was seen in the cross-comparison of the three xenoestrogens and GSI. Upon a cross examination, we saw that EE₂ was the most potent of the three chemicals, displaying the largest effects at the lowest concentration.

Introduction

Human activity has resulted in toxic, chemical runoff being thrown into water. This pollution could be from industrial byproducts (Clara *et al.* 2010) or from agricultural practices.

One particular toxic category of chemicals found in water is called endocrine disruptors. In order to understand what these are, we first have to define the endocrine system, which is essentially the system of glands that produce hormones and release them into the bloodstream for various types of regulation. One particular group of endocrine disruptors that has come under recent scrutiny from the scientific community is xenoestrogens—exogenous chemicals that cause unintended estrogenic activity in vertebrates. Three vertebrate estrogenic receptors are known to be highly promiscuous and are able to interact with a variety of molecules other than their intended hormone (Brevini *et al.* 2005). Exposure to xenoestrogens has been shown to feminize male fish or reverse their sex entirely (Uren-Webster *et al.* 2010). In some cases, intersex fish have been observed (Norman *et al.* 2005).

Three xenoestrogens in particular were studied for the purpose of this analysis: phthalates, 4-alkylphenols, and 17 α -ethynylestradiol (EE₂). Phthalates are synthetic compounds that are often used as plastic additives and are used globally, particularly in Europe and North America (Clara *et al.* 2010). Due to their wide use in industry, there has been global concerns about the ecotoxicological effect phthalates could have on different species, particularly marine species (Hemming *et al.* 2001). Analyses on storm water runoff and tribute waters show that different phthalates are leaking into the environment and as a result, fish demographics have been altered in some areas (Börkland *et al.* 2009, Harries *et al.* 2000).

Alkylphenols are a class of commercially important precursors for industrial detergents that are known environmental pollutants (Zoller 2006). The vast majority of the studied compounds in the literature focus on the para forms of structurally similar compounds nonylphenol and octylphenol, where the hydroxyl group is located in a 1,4 relation to the alkyl

group on a benzene ring. Alkylphenols have been shown by numerous studies to induce estrogenic activity in a wide variety of fish species (Soverchia *et al.* 2005, Seki *et al.* 2003, Hemmer *et al.* 2000).

17 α -ethynylestradiol (EE₂) is a synthetic form of estrogen used primarily in pharmaceutical treatments such as oral contraceptives, hormone replacement therapy for menopause and osteoporosis, breast and prostate cancer therapy, and veterinary medicine used for growth promotion (Wise *et al.* 2011). Wise *et al.* 2011 at the University of California, San Francisco claim that 40% of EE₂ that is ingested by humans reaches sewage as available EE₂. This sewage is then dispersed into aquatic environments through industrial and agricultural runoff, thus, acting as a xenoestrogen to aquatic life (Folmar *et al.* 2002).

The effects of the exposure to the three xenoestrogens can be observed on two biomarkers: the presence or change in the concentration of vitellogenin (VTG) and the change in gonadosomatic index (GSI). VTG is a protein that is made predominantly during egg-yolk formation in female fish (Kim *et al.* 2002). Normally only found in considerable amounts in female fish, trends of VTG levels increasing in males serves as an indicator as to whether the experimental chemicals exhibit xenoestrogenic effects (Kim *et al.* 2002). GSI is a measure of gonad weight over body weight as a percentage, and is a good physiological indicator of xenoestrogenic effect due to contaminants in the water (Kim *et al.* 2002). If values of GSI in male fish decrease, it is an indication of retardation in gonad development and feminization.

The purpose of this meta-analysis is two-fold. First, we will analyze experimental data that test the effects of different mass concentrations of phthalates, 4-alkylphenols, and 17 α -ethynylestradiol on the two aforementioned biomarkers--VTG and GSI. Then, we will take the

experimental results of the effects on these biomarkers and attempt to discern which of these pollutants is the most harmful to aquatic ecology by doing a cross-comparison statistical test on low concentrations of each chemical. In other words, this will tell us which chemical is the most potent, and thus, the most harmful to fish. Based on preliminary data, we hypothesize that increasing the concentrations of each chemical will lead to increased values of VTG concentration and decreased values of GSI in male fish; from preliminary research, we also expect EE₂ to be the strongest chemical factor of the three xenoestrogens in the feminization of fish.

Methods

Obtaining Articles

Several Internet databases such as Pubmed, Web of Knowledge and Biological Abstract were used in order to search and obtain studies relevant to this meta-analysis. Keywords such as “DEHP,” “vitellogenin,” “gonadosomatic index,” “4-octylphenol”, “17 α -ethynylestradiol,” and “feminization” were used in order to select studies that fit the targeted criteria. EndnoteWeb 3.5 provided by Web of Knowledge was used to consolidate and organize studies into useful groups such as “Phthalate Group” or “Estrogen Group”. EndnoteWeb also provided a function that allowed access to full text of desired studies.

Criteria of Articles

In order to obtain comparable data on the effects of phthalates, alkylphenols, and EE₂, certain criteria were established for selecting appropriate studies in order to reduce the effects of confounding variables. The subjects analyzed were always chordates, based on the assumption that all subjects have a similar ancestral endocrine system (Sower *et al.* 2009); their responses to

the different xenoestrogens can be generalized to other chordates. The subjects used were the Chinese rare minnow (*Gobiocypris rarus*), common carp (*Cyprinus carpio*), eelpout (*Zoarces viviparus*), fathead minnow (*Pimephales promelas*), goldfish (*Carassius auratus auratus*), guppy (*Poecilia reticulata*), Japanese medaka (*Oryzias latipes*), rainbow trout (*Oncorhynchus mykiss*), sheepshead minnow (*Cyprinodon variegatus*), and zebrafish (*Danio rerio*). For all three xenoestrogens studied, the age of interest in the experiments were fish in the adult phase in order to eliminate variability in the results; however due to lack of available research, the fish age range varied for the phthalate study from eggs to adult fish.

All data came from essentially the same experimental set-ups throughout all the studies that were researched. Male fish were obtained from a fish hatchery and placed in tanks of water with varying concentrations of the particular xenoestrogen chosen by the experimenters, and data was collected after given periods of time. The data of interest in particular was the effects of the xenoestrogens in concentrations on levels of VTG and the GSI values after exposure to the chemicals for at least 4-28 days, with preference for a period of at least two weeks. This range was arbitrarily chosen because it was typically included in the experiments that were researched for all three chemicals, and isolating data in a specific duration range better eliminated variability in the results. The results for VTG and GSI obtained from each experiment were recorded on a table with their corresponding dosages (Table 1, 2, 3, 4, and 5). Nearly all individual experiments for all three xenoestrogens conducted more than one trial for each test concentration, and the resulting VTG and GSI values were given as an average for the trials with a reported standard error. For the purpose of our experiment, the standard errors recorded in the studies were ignored

and only the averages were recorded in our raw data. Thus, we will be testing under the assumption that the averages recorded for those trials do not include the standard error.

Statistical Analysis

Once all the data from the experiments were recorded, each chemical was independently assigned a concentration range that was considered low, medium, or high based on overall concentrations used by the experimenters for each chemical; the control was always a concentration of 0 $\mu\text{g/L}$ for all three chemicals as that indicated no chemical was added. The concentration ranges for phthalates were set based on the data from the concentrations used in phthalate studies only. The low concentration range was from 0 to 1 $\mu\text{g/L}$, the medium range was 1 to 50 $\mu\text{g/L}$ and the large concentration range was 50 $\mu\text{g/L}$ or greater. Based on the distribution of data from selected studies, concentration ranges for alkylphenols were separated by 10, 50, and 100 $\mu\text{g/L}$; values between 0 and 10 $\mu\text{g/L}$ were assigned to the low range, those between 10 and 50 $\mu\text{g/L}$ to the medium range, and those between 50 to 100 $\mu\text{g/L}$ to the high range. Since the data for EE₂ was almost entirely measured to values that were in the degrees of 10^{-3} , 10^{-2} , or 10^{-1} $\mu\text{g/L}$, any value that was greater than 0 $\mu\text{g/L}$ and measured in a degree of 10^{-3} $\mu\text{g/L}$ was assigned to a low range; any value in the degree of 10^{-2} $\mu\text{g/L}$ was assigned to a medium range; and any value of in the degree 10^{-1} $\mu\text{g/L}$ was categorized as high. Based on the data obtained, the exact ranges for EE₂ were established as the following: greater than 0 $\mu\text{g/L}$ to 0.005 $\mu\text{g/L}$ was low; 0.01 $\mu\text{g/L}$ to 0.04 $\mu\text{g/L}$ was medium; and 0.1 $\mu\text{g/L}$ or greater was considered high. According to Zha, *et al.* 2007, the average nominal concentrations of EE₂ in aquatic environments is reported to be 0.015 $\mu\text{g/L}$, which appropriately falls into the medium range established for EE₂ and

validates the three ranges established. Since these ranges were designated after all the research data was gathered, all data for each chemical falls in at least one of the three ranges for EE₂.

Two ANOVA tests were run separately for each chemical. One ANOVA test was generated for comparing the control, low, medium, and high ranges against concentration of VTG measured in mg/ml, and a separate ANOVA test compared the same ranges and their corresponding GSI values, giving a total of six ANOVA tests. These tests compared the variation between the averages of the control and the three ranges based on a significance level of $\alpha = 0.05$.

Any ANOVA test with a p-value below 0.05 meant that there was less than a 5% chance that the result was due to random chance; therefore, there was a 95% confidence level that our results could be reproduced consistently, indicating a statistically significant value. However, p-values above 0.05 may still have reflected trends in the relationship between the concentrations ranges and the corresponding VTG and GSI values.

In order to determine which chemical was the most harmful in low concentrations, two more ANOVA tests were conducted using the low-range concentrations of each chemical. Similar to the previous ANOVA tests, one compared VTG concentration levels and one compared the GSI values. For these two ANOVA tests, the same significance level of 0.05 was used to determine whether change in the treatment of the chemical was statistically significant to VTG concentration and GSI values. Furthermore, the average value of the VTG and GSI were observed at the lowest concentrations to determine which chemical had the strongest effects.

Results

Phthalates

The ANOVA test comparing averages between VTG concentration and the concentration ranges of phthalates used three species of fish: the common carp, fathead minnow and zebrafish. The studies ranged from 4-28 days of exposure to diethyl phthalate (DEP) and di-2-ethylhexyl phthalate (DEHP).

The VTG concentration for the control group was 0.000067 ± 0.000057 mg/ml . Comparatively, the average VTG concentration for the low dosage of phthalate exposure was 0.000001 ± 0 mg/ml , the medium dosage had an average of 0.000004 ± 0.000001 mg/ml and the high dosage had an average of 0.000269 ± 0.000093 mg/ml (**Figure 1**).

Only the high concentration of phthalate resulted in an increase in VTG compared to the control; however, the VTG concentration was five times greater than the control. The low and medium concentrations showed a decrease in VTG concentration. Overall, the p-value for VTG concentration and different phthalate concentration ranges was 0.032, which is statistically significant.

The data used for the ANOVA test comparing GSI and phthalate concentration used two species of fish, the Japanese Medaka and fathead minnow. The duration of exposure these studies experimented were 12-21 days and the age of fish used in this data set ranged from egg to adult. The phthalates tested were DEHP and Butyl Benzyl Phthalate (BBP). The ANOVA test showed a general increase in GSI value with increased phthalate concentration and a p-value of 0.000169, which is statistically significant.

The control group resulted in a GSI of 0.91 ± 0.06 %. The average GSI for the low dosage was 0.63 ± 0.061779 %, the GSI for the medium dosage was 0.68 ± 0.039158 % and the

GSI for the high dosage was 1.7%. The high dosage does not have a standard of error because only one value was used for this ANOVA test (**Figure 2**).

Alkylphenols

Table 3 displays the combined results from eight experiments with NP (nonylphenol) or OP (octylphenol). The data demonstrate a dose-dependent increase in VTG from exposure to increasing concentrations of alkylphenols, and a corresponding reduction in GSI percent. Control males showed a mean VTG induction of 0.001009 ± 0.000999 mg/ml. While only the high dose showed a large increase in VTG concentration, with a mean of 4.833537 ± 2.574371 mg/ml (over 2000 to 4000 times that of the control), the other dose ranges only yielded VTG concentrations of 0.00864 ± 0.003426 mg/ml (low) and 0.005718 ± 0.005536 mg/ml (medium). These latter two average responses, while comparatively modest in increase, still are roughly eight and five times the mean amount of VTG observed in the controls. Results from the ANOVA test on all VTG data included a p-value of 0.000871, which is lower than (and thus statistically significant at) our chosen alpha level of $\alpha = 0.05$.

Table 4 presents data collected from three experiments that measured GSI in fish test subjects after exposure to alkylphenols. The mean GSI in controls was 1.3875 ± 0.141973 %. The low and medium dose ranges resulted in higher average GSI responses of 2.425 ± 0.749861 % and 4.9 %, respectively. The medium range had too few values ($n = 1$) to report a standard error. However, the high dose range yielded an average GSI of 0.666667 ± 0.218581 % in test subjects, or about half that of the control. ANOVA results for all dose ranges gave a p-value of 0.021413, which is also statistically significant at $\alpha = 0.05$. **Figure 3** offers a graphical

representation of the mean VTG and GSI levels for each dose range, with standard errors reported.

EE₂

Only two different species were used for EE₂—zebrafish and fathead minnows. The durations of the experiments were in the range of 12-21 days. **Table 5** shows the data obtained from five different experiments for EE₂. Based on the data of the five experiments individually, there seems to be an increase in the concentration of plasma VTG as the concentration of EE₂ increased. Inversely, there is generally a decrease in GSI value with higher exposure to EE₂. The ANOVA test for the three ranges of EE₂ concentrations and their corresponding VTG concentrations resulted in a p-value of about 0.07, which by our chosen level of significance, is not considered to be significant. The ANOVA conducted similarly for the GSI gave a p-value of 0.02--a significant value.

The EE₂ control group and the three concentration ranges and their respective averages for VTG concentration and GSI are shown numerically under the graph in **Figure 4** with their standard errors. The average of the VTG levels increased from the lowest range to the highest range. The average for controls (0 µg/L EE₂) was 1.22 ± 0.8026 mg/ml, followed by 2.81 ± 1.8239 mg/ml for the low range (0.0001 to 0.005 µg/L of EE₂), 18.60 ± 5.4876 mg/ml for the medium range (0.01 to 0.04 µg/L of EE₂), and 22.04 ± 21.9635 for the high range (0.01 µg/L of EE₂). There was a decrease in the average GSI percent as the concentration range went from low to high. The control had a GSI of 1.1186 ± 0.202059 %, followed 1.0106 ± 0.104242 % for the low EE₂ concentration range, 0.680556 ± 0.062652 % for the medium range, and 0.505 ± 0.005 %for the high range.

Similar to the individual results of each EE₂ experiment, there was a correlation with increasing average levels of VTG and decreasing average levels of GSI as the concentration range of EE₂ increased. It should be noted, however, that after the individual EE₂ experimental data was observed, and the ranges for low, medium, and high were established, the low range consisted of five total values, the medium range consisted of nine total values, and the high range only consisted of two total values as shown in **Table 6**.

Cross-Comparing the Lowest Concentrations

After comparing the lowest concentrations of each chemical through an ANOVA test for VTG and GSI, the p-values were found to be 0.02 and 0.03 respectively. These values indicate levels of significance that are considered to be statistically significant based on our chosen value of $\alpha = 0.05$.

The average concentration of VTG for fish exposed to EE₂ was 2.809517 ± 1.823941 mg/ml, fish exposed to alkylphenols had an average of 0.00864 ± 0.003426 mg/ml and phthalate exposure resulted in an average of 0.000001 ± 0 mg/ml of VTG. On the other hand, the GSI value for fish exposed to EE₂ was 1.0106 ± 0.104242 %, alkylphenols produced an average GSI value of 2.425 ± 0.749861 % and phthalate exposure resulted in an average of 0.63 ± 0.061779 % for GSI.

The lowest concentration ranges and their respective averages for VTG and GSI are shown graphically and numerically in **Figure 5**. At the lowest concentrations, VTG levels seem to be highest for EE₂, followed by alkylphenol and phthalates. No apparent correlation was seen for GSI based on the cross-comparison of the lowest concentration range.

Discussion

Phthalates

The results obtained from the ANOVA test between VTG concentrations and phthalate exposure showed no obvious trends, but showed that the differences in VTG concentrations were statistically significant among the four concentrations. From **Figure 1**, it was observed that an increase in phthalate concentration above 50 µg/L would result in high VTG induction in male fish. This result is similar to those of Maradonna *et al.* 2012 and Barse *et al.* 2007 and supports the hypothesis that phthalates do have estrogenic effects (**Table 1**). Uren-Webster *et al.* 2010 also observed a significant increase in VTG concentration with exposure to high concentrations of phthalates, specifically DEHP. Atlantic Salmon exposed to 5000 mg DEHP/ kg of fish body weight resulted in nearly a four-fold increase in VTG concentration compared to the control (Uren-Webster *et al.* 2010).

The p-value of 0.032 gives us a 95% confidence to fail to reject the hypothesis that phthalates do have estrogenic effects on fish. Based on the results, the effects of the phthalates were only noticeable at high concentrations on VTG induction, which indicate that phthalates concentrations below 50 µg/L are not potent enough to induce vitellogenin production in males. Uren-Webster *et al.* 2010 also noted that in order to induce vitellogenin transcription in male fish, unnaturally large concentrations of phthalates must be present in the environment.

Another important note is that the strength of this ANOVA is not statistically strong due to low sample sizes. The control and low concentration had a sample size of three, the medium range had a sample size of two and the high range had a sample size of four. Because the sample sizes for each range were small, the data from the ANOVA must be interpreted with caution. In order to run a stronger ANOVA test, more studies would have had to be included in order to

generate more reliable averages and reduce the effects of outliers. However, the data presented by the ANOVA is consistent with data collected by Barse *et al.* 2007, Maradonna *et al.* 2012, and Uren-Webster *et al.* 2010, leading to the conclusion that phthalates have the ability to increase VTG production, a trait of feminization.

One factor not included in this meta-analysis was measuring phthalate concentration affecting fish by means other than water contamination. Studies have been conducted where subjects have been injected with doses of phthalates in respect to the subject's weight (Uren-Webster *et al.* 2010 and Christiansen *et al.* 1998). VTG induction has been seen by fish treated with food exposed to differing concentrations of phthalates (Uren-Webster *et al.* 2010). A more comprehensive study that measured more than one method of chemical exposure could lead to more definitive trends in regard to phthalate potency.

The ANOVA test comparing GSI values and phthalate concentrations showed no statistically significant differences among concentrations, but a slight positive trend was noticed. As tested phthalate concentrations increased, the average GSI values increased. The ANOVA test displayed a positive trend between GSI and phthalate concentration, resulting in confounding data. The positive trend was unexpected because previous studies have shown that GSI tends to decrease with increased exposure to endocrine disruptors (Kim *et al.* 2002).

The positive association between GSI and phthalate concentration from this meta-analysis are counterintuitive because phthalates have been seen to show emasculating effects to a wide array of male fish (Barse *et al.* 2007). All studies besides Harries *et al.* 2000 observed a decrease in GSI compared to the control groups. However Harries *et al.* 2000 did not see any significant change in gonad size during exposure to Butyl Benzyl Phthalate (BBP), which

indicated that the results from Harries *et al.* 2000 could be an outlier in this meta-analysis. If the data collected from Harries *et al.* 2000 were not included in this meta-analysis, the association between GSI value and phthalates would show a negative correlation. A negative correlation would be consistent with data collected from other studies exposing fish to phthalates, as well as other endocrine disrupting chemicals. The impact of the data from Harries *et al.* 2000 is so crucial because that study provided the only data value for GSI at the high concentration of phthalate exposure, another example of the crippling effect of small sample sizes.

The p-value of 0.000169 is statistically significant in determining differences among varying phthalate concentrations and GSI values. The increasing values for GSI with increasing concentration of phthalates can be partly attributed to the small sample size. The sample size for control and low concentration were four, the medium concentration had a sample size of six and the large concentration only had a sample size of one. The standard error for the large concentration was zero due to the limited sample size.

A larger sample size for each concentration should be taken to reduce the effects of outliers from the data as well as enforce the law of large numbers. The possibility for high variance among samples is large and could not be corrected for in this meta-analysis because of the limited research available. Ideally, a large enough sample size would be able to produce more expected results while minimizing the effect outliers have on the average.

A possible reason for the difference between this meta-analysis and previous studies regarding phthalates was the meta-analysis included studies that varied in time of exposure. Chickae *et al.* 2004 exposed Japanese medaka eggs between 12 to 15 days before hatching was observed, whereas Chikae *et al.* 2003 exposed medaka to different phthalate concentrations for

21 days. The duration of exposure could be an important factor in determining magnitude of effect in fish.

Because phthalates have been identified as having weaker estrogenic effects than other xenoestrogens like 17 β -estradiol (Harries *et al.* 2000), the effects of phthalates are expected to be less noticeable in the same time frame compared to other xenoestrogens. Therefore, a more representative study would include a smaller range of time exposure for comparing different chemicals. However, due to the limited number of studies involving the effects of phthalates on VTG and GSI in fish, such a large time interval had to be chosen for this meta-analysis.

Another possible confounding variable was the age at which the fish were tested. The subjects tested for GSI were typically in the fry stage of the life cycle, and therefore had not yet matured. The effects of DEHP impaired body growth of male medaka after exposure (Chikae *et al.* 2004). Because body growth was hindered and the gonads had not changed much throughout the testing period, it is reasonable to deduce that smaller than expected GSI values were recorded from Chikae *et al.* 2004.

Ultimately, the exact mechanisms of phthalate interaction in fish has yet to be fully explored and cannot be determined from this meta-analysis. However, there has been speculation on how phthalates affect the biochemical pathways of fish. The induction of VTG is most likely due to the phthalate chemical activating an estrogen receptor in fish, signaling VTG transcription (Christiansen *et al.* 1998). On the other hand, phthalates, most notably DEHP, has displayed anti-androgenic effects on fish, resulting in the reduction in testicular growth and development and in effect, decreasing GSI values (Chikae *et al.* 2004).

Alkylphenols

Examination of raw data indicated that the alkylphenols NP and OP, when viewed together, had a strong positive correlation with VTG uptake, and a weaker negative correlation with GSI percent. The ANOVA tests also confirmed these correlations and their statistical significance; with 95% confidence we fail to reject the hypotheses that increased exposure to alkylphenols leads to estrogenic effects in fish, as measured by VTG and GSI levels.

As shown in **Figure 3**, the high dose range had a dramatically larger effect on VTG induction than the lower dose ranges, to an extent that their means were overshadowed on the graph. With a mean induction of 4.8335737 ± 2.574371 mg/ml, high doses of alkylphenols are shown to be linked with higher VTG concentrations in subjects. Even taking the large standard error into account, high doses still showed a formidable impact on VTG. Lower doses produced comparatively negligible results, and even on the high ends of their standard errors, the low and medium dose ranges barely exceeded 0.01 mg/ml VTG. The p-value for VTG data was very low, at 0.00087. For our tested 95% confidence level at $\alpha = 0.05$, such a result is highly unlikely to be explained by random chance, and even at much higher confidence levels, it would still be statistically significant. Thus the data suggest that alkylphenols do have a major estrogenic effect that increases VTG uptake with elevated treatment doses, a finding consistent with Hemmer *et al.* 2000, who reported “a dose-dependent increase of plasma VTG over the entire time of exposure, with significantly elevated VTG levels by the fifth day of exposure to *p*-nonylphenol at concentrations of 5.4 µg/L or greater” in the minnow *Cyprinodon variegatus* (Hemmer *et al.* 2000). A similar study by Zhang *et al.* 2007 found significant *vtg* mRNA expression increase in male *Gobiocypris rarus* liver tissue after 21 days of exposure to 2 µg/L NP or greater, offering epigenetic evidence of alkylphenol endocrine disruption (Zhang *et al.* 2007). Because our meta-

analysis only examined VTG uptake in a total of 39 treatment groups, additional data for each alkylphenol dose range are necessary supplement and confirm our limited findings, which may suffer from sampling bias. Additionally, 14 of those groups were from Seki *et al.* 2003, which observed a combined sample of juvenile and adult fish. Although our criteria specified that subjects should be adult males, this study was allowed on the basis that all other criteria were met, and observed results were consistent with VTG values, across dosage ranges, from other studies.

The p-value for GSI data, 0.021413, is also statistically significant at 95 % confidence, but is considerably closer to our α level, which decreases its practical confidence overall. Though the mean GSI % of each dose range did not always decrease with an increased dose of alkylphenols, test results still support a negative correlation between GSI and treatment dose range. It is probable that our tests suffered from sampling bias, due to limited availability of studies on alkylphenol xenoestrogenic activity that used GSI as a metric (12 sample groups). Many GSI-orientated studies focused solely on growing embryos, or fish that had been directly injected with alkylphenols, and were unavailable for inclusion in our meta-analysis. Only a single value fell into the medium range for GSI, and it was the highest observed percentage in the dataset (4.9 %). Therefore, a standard error was unfortunately unobtainable for the medium range. Because the sole medium data point was twice the value of the low dose range's mean, the data is counterintuitive; as more estrogens are present, gonads are expected to be feminized, and their percent contribution to body weight decreased. Likely, the point was an outlier, and more data in the medium range should provide evidence for a stronger negative correlation between alkylphenol concentration and GSI. Nonetheless, our results are consistent with findings from

Christiansen *et al.* 1997 that report a statistically significant increase in GSI in *Zoarces viviparus* after 21 days of treatment with either 1 or 100 µg/L NP. If GSI percent is modulated by alkylphenol exposure, many more data points will be necessary before estrogenic effects can be demonstrated with certainty.

Because the pool of available alkylphenol studies that fit our selection criteria was limited, much of the data comes from fish species that differ widely, from goldfish to minnows. Variability between fish species is a potential confounder of the data because some species may be inherently larger or more resistant than others to toxic or estrogenic effects from similar doses. Ideally, dosages would be set to units of µg/kg body weight, to control for differences, subtle or otherwise, in fish anatomy. However, because not all studies weighed their test subjects, such a metric was unobtainable for our analysis. Differences in exposure length between studies could also confound results, as the links between VTG induction or GSI % and alkylphenol exposure are time-dependent (Harries *et al.* 2000). Nonylphenol and Octylphenol were not compared to each other, based on the assumption that their high level of structural similarity would lead to similar responses in test subjects. Although Laws *et al.* 2000 found the two chemicals exhibited comparatively weak yet equal binding affinities for the progesterone receptors in rats, no such comparison has been tested in fish, and potential variability in their effects on aquatic species may have lead to confounding of data (Laws *et al.* 2000). Nonylphenol was overwhelmingly preferred by researchers as a test chemical, however, and any existing deviation from expected octylphenol responses would likely be low enough to have a minimal impact on the validity of our results.

For practical purposes, our findings would be more applicable to the field of xenoestrogen toxicology if they could be compared to known environmental concentrations of alkylphenol pollutants. Since pollution is not uniformly distributed among waterways and thus not among fish species, results would have to be assayed against observed concentrations in fishes' natural habitats to attain maximum accuracy (Hemming *et al.* 2001).

EE₂

Based on the raw data of the five figures alone, increasing EE₂ concentration seemed to have a clear trend for both VTG and GSI. Each experiment shown in **Table 5** generally showed an increase in plasma VTG and a decrease in GSI as EE₂ concentrations increased. The same trend was found when the experimental data in **Table 5** was categorized into the three ranges, and the average values for VTG and GSI were calculated. After running the two ANOVA tests for the concentration ranges of EE₂ versus VTG and GSI, the p-value of 0.07 for the VTG test was not found to be statistically significant. However, a significant corresponding p-value of 0.05 is only arbitrarily chosen. The 0.07 is close enough to 0.05 that we can still infer there is a correlation with higher levels of VTG in the presence of higher concentrations of EE₂. The ANOVA test for the GSI values yielded a p-value of 0.02 for the three concentration ranges of EE₂, and this value indicated statistical significance. In other words, the correlation here was that as EE₂ concentrations increased from lowest to highest, the GSI values were most likely to decrease.

However, there is something important that needs to be accounted for. When looking at the three ranges of the chemical concentrations that were assigned as low, medium, or high, the number of values in each category were different as mentioned previously (**Table 6**).

The fact that the low concentration range only consisted of five values, the medium concentration range consisted of nine values, and the high concentration range only had two values implies that the results in the ANOVA tests may not be entirely reliable. For example, if we look at the standard errors for the average VTG levels in the medium EE₂ concentration range, compared to the low and high ranges, the medium range had the smallest standard error relative to the average. This is because the medium range consisted of a total of nine values from the raw data shown in **Table 5**, as opposed to five in the low range and two in the high range. If we consider the high concentration range for EE₂, both concentration values for EE₂ were 0.1 µg/L. However, one experimented with zebrafish and the other experimented with fathead minnow. The resulting VTG values were different by a factor of 602, even though both species were under 0.1 µg/L of EE₂. This also explains why the standard error for the average VTG concentration in the high EE₂ range was almost equal to the average itself; it was due to a combination of having only two data points in that range as well as differing species. Perhaps in future experiments, the only way this problem can be addressed is by doing additional research for more data points in a particular range, as well as having consistency in the species tested.

Relating to the point made previously, one of the biggest sources of variability for testing exposure to EE₂ was the varying fish species. In Van Den Belt *et al.* 2001, Van Den Belt *et al.* 2002, and Versonnen *et al.* 2004, the species of fish used was Zebrafish while the Pawlowski *et al.* 2004 and Salierno *et al.* 2009 conducted their experiments on fathead minnows. To reiterate mathematically why differences is significant, we can again compare two data points that tested the same concentration on two different species—one from the 2001 Van den Belt experiment to one from Pawlowski's 2004 experiment. At a concentration of 0.01 µg/L, the Van

den Belt experiment had a VTG value that is approximately 496 times greater. Thus, differences in the species was certainly a source of variability based on the methods we used.

Another source of variability in the results of EE₂ stemmed from the duration of the experiments with regards to exposure to EE₂. If we look at the experimental data on Van Den Belt, *et al.* 2002 in **Table 5**, the results reported on the data only showed the resulting VTG and GSI values after twelve days of exposure. The values after 12 days were chosen because it was almost in the middle of our target range of 4-28 days of exposure. But if we look at the full data presented in the experiment, there was a rather different trend observed at day 24 of the experiment. It turned out that the concentration of VTG for the EE₂ concentration of 0.01 µg/L increased from 22.5 mg/ml on day 12 to 32.5 µg/L on day 24 as expected, but when the concentration of EE₂ was increased to 0.025 µg/L, the VTG levels for 12 days exposure was 30 mg/ml while the VTG levels for 24 days of exposure decreased to 25 mg/ml. In other words, this experiment showed that for a certain number of days of exposure, there was indeed a direct correlation with increasing concentrations of EE₂ resulting in increased VTG concentrations. After a certain number of days however, the VTG values seemed to decline. Therefore, if the data point of 24 days were used for this experiment, this would have affected the ANOVA test, and thus, the correlations that were reported in our results with regard to VTG. So there may be differing results based on the duration of the experiment. However, it is not to say that higher EE₂ doesn't have as strong of an effect for 24 days as it does for 12 days. Van Den Belt, *et al.* 2002 also reported that for the higher EE₂ concentration group, the rate at which VTG increased was much faster than for the lower concentrations, even though the higher concentration group hit a peak. Furthermore, when the fish were allowed to recover in water containing no EE₂ after

the full duration of the EE₂ exposure period, the fish exposed to higher concentration of EE₂ for 24 days took longer for their VTG levels to return to normal as opposed to fish exposed to lower levels of EE₂ for 24 days. Thus, increased EE₂ concentrations do have an effect in the long term in the VTG levels of fish.

Comparing the Three Xenoestrogens

Before the ANOVA tests were conducted for cross-comparison, we hypothesized that EE₂ would be the strongest chemical because all experimental data found for EE₂ had chemical concentrations in units of nanograms/liters (ng/L). This is a difference by a factor of 10⁻³ in units measured, compared to alkylphenols and phthalates, which were measured in µg/L. Thus, this was an early indicator that EE₂, although present in smaller concentrations, was more harmful to fish than the other xenoestrogens.

According to Maradonna, *et al.* 2012, it took about a ten times higher increase in DEHP concentration compared to EE₂ to have similar effects on VTG induction of Zebrafish. Another experiment by Zha, *et al.* 2007 indicated that alkylphenol concentration had to be 3000 times higher than the concentration of EE₂ to induce the same level of VTG on Rare Minnow, and about 2000 times the concentration of EE₂ to induce the same decrease GSI. Results of a third study conducted by Harries, *et al.* 2000 showed that at the same concentration of OP (an alkylphenol) and BBP (a phthalate) showed that VTG production was about 1000 times greater in Fathead minnows exposed to OP. From the information reported in these three studies, it can be inferred that EE₂ was the most potent xenoestrogen, alkylphenols was second, and phthalates were the least potent of the three xenoestrogens.

After performing the two ANOVA tests for the lowest concentrations of each chemical with respect to VTG and GSI, we obtained statistical evidence for differences in estrogenic effects between our three examined classes of chemicals. **Figure 5** shows the average VTG and GSI responses for the lowest concentration range of each xenoestrogen. For EE₂ in low concentrations, there seemed to be a significant increase in the levels of VTG present in male fish compared to alkylphenols, even when standard error was taken into account. The p-value for this test was 0.02, indicating that this result was statistically significant at a level of 95% confidence. This p-value indicates that the three chemicals indeed have different potencies in regard to VTG production. The trend shown in the **Figure 5** clearly supports our hypothesis that EE₂ was the strongest inducer of VTG, followed by alkylphenols and phthalates.

The GSI ANOVA also had a significant p-value of 0.03, but average GSI values for the lowest concentration range of each chemical did not follow the trend expected. Experimental results generally showed an inverse relationship with levels of VTG and GSI; thus, if VTG levels were highest in EE₂ followed by alkylphenols and phthalates, we would have expected to see lowest levels in GSI of EE₂, followed again by alkylphenols and phthalates. However, the results showed that phthalates had the lowest average GSI in the low concentration range. Such a problem in accuracy likely reflects the GSI response values for phthalates, which were taken from experiments conducted on immature fish rather than adult fish; again this was due to lack of available research on phthalates on adult male fish.

Based on the analysis of the data from the various experiments, it is evident that increasing the concentrations of each xenoestrogen has an effect on both biomarkers; however, the lack of available data for phthalates did not allow us to correctly prove our hypothesis that it

would decrease the values in GSI. More research in this field is necessary to accurately prove or disprove this claim. Overall, the biomarker that was certainly affected was VTG, and our data supports the idea that increasing the concentrations of the xenoestrogens increases the concentrations of VTG in male fish.

Fortunately, some studies took their experiments further by placing the fish that were initially exposed to xenoestrogens in a new environment where xenoestrogens were no longer present. The results showed that the effects caused by xenoestrogens could be reversed by varying degrees. For example, Norman *et al.* 2005 found that for the phthalates DEHP and MEHP, both were eliminated from Atlantic Salmon one week after being exposed to those chemicals, however, the results were irreversible in that inter-sex fish were observed. Seki *et al.* 2003 reported that in Japanese medaka exposed to alkylphenols, secondary sex characteristics such as VTG reverted to normal after a return to clean water, but intersex gonads remained in test subjects after two months, suggesting the potential for incomplete reversibility of alkylphenols' xenoestrogenic effects. The Van Den Belt *et al.* 2002 experiment for EE₂ on Zebrafish showed that after a 24-day recovery period where no EE₂ was present, fish that were initially exposed to EE₂ for 24 days completely recovered in terms of gonad size, indicating that effects on GSI were reversible for Zebrafish.

Based on the information for the recovery of fish presented by the previous studies, it is possible to reverse some of the effects of xenoestrogens on fish if current practices of chemical disposal are improved. Sewage plants can modify their methods of screening for chemical pollutants to include known xenoestrogens, and filter them out of effluent before they enter aquatic ecosystems. "Existing wastewater treatment technologies have the potential to eliminate

most of the estrogenic and androgenic activity before discharge to the environment” (Eggen *et al.* 2003). Because affected fish often are unable to revert from their induced intersex states, populations in contaminated water may be in danger of collapse if current dumping practices continue. The lasting effects of xenoestrogens on human health are still under debate and poorly understood, highlighting the need for further research on bioaccumulation of estrogens in fisheries harvested and maintained by people.

Tables and Figures

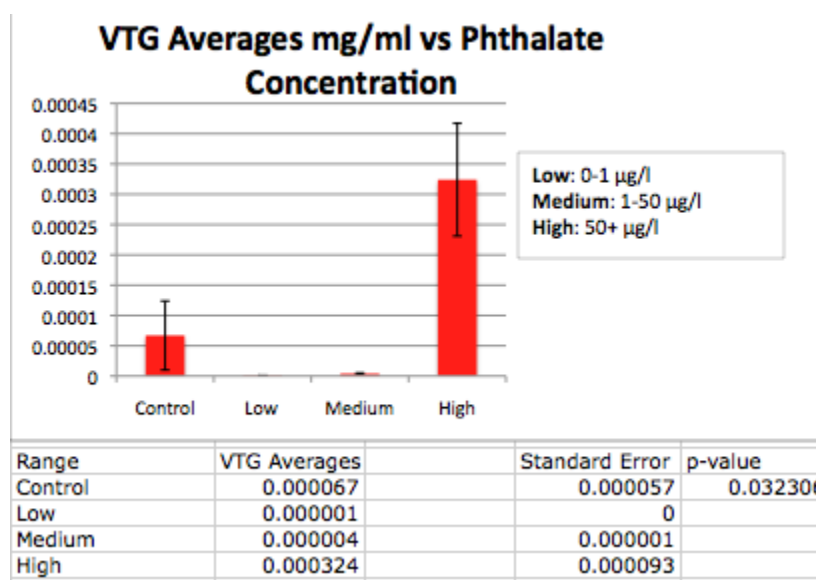


Figure 1. The averages of VTG concentration for the experimental concentration ranges for phthalates with standard errors. Only VTG levels in fish exposed to phthalate concentrations of more than 50 µg/L expressed a considerable increase.

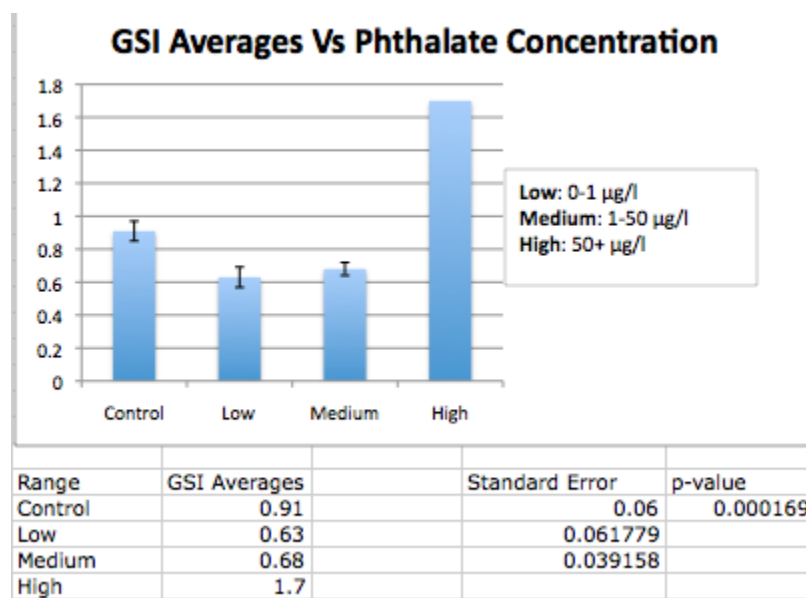


Figure 2. GSI values at given phthalate concentration ranges with standards of error. An increase in GSI was observed with increasing concentration of phthalates with the high concentration displaying the largest GSI value.

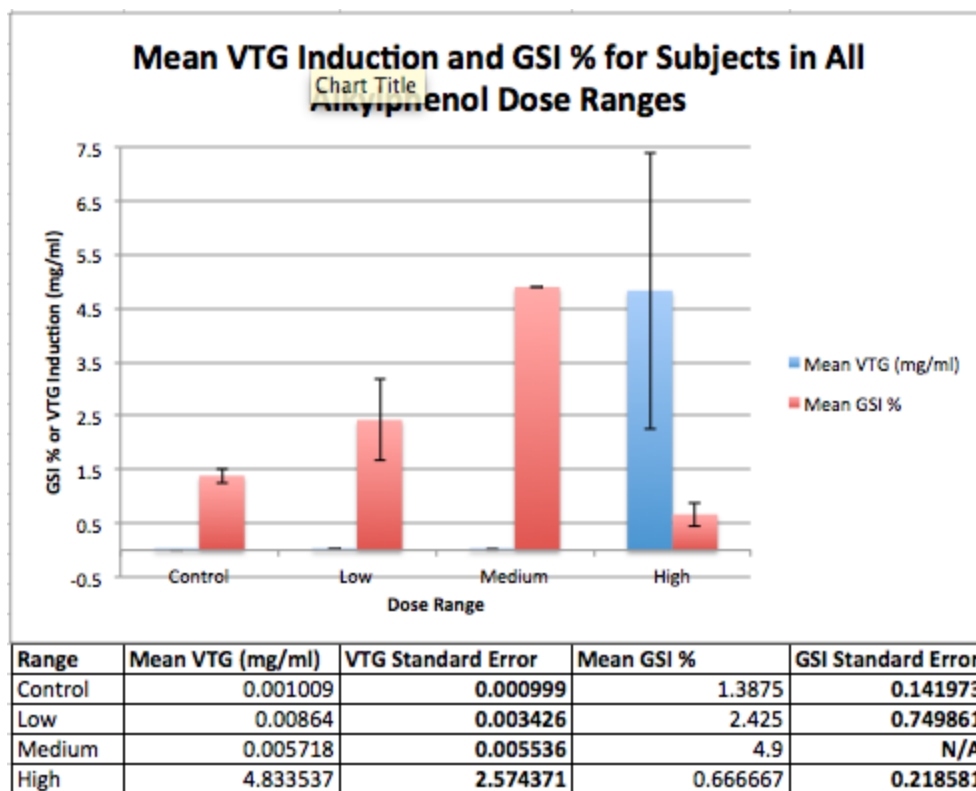


Figure 3 shows the mean GSI and VTG values in subjects for each dose range, with standard errors. While the figure appears to suggest GSI % increases with dose, statistical results determined an overall negative correlation. As the mean VTG level for the high dose range was far higher than the others, it is the only VTG bar clearly visible on the figure. The medium range of GSI had too few data values to report a standard error.

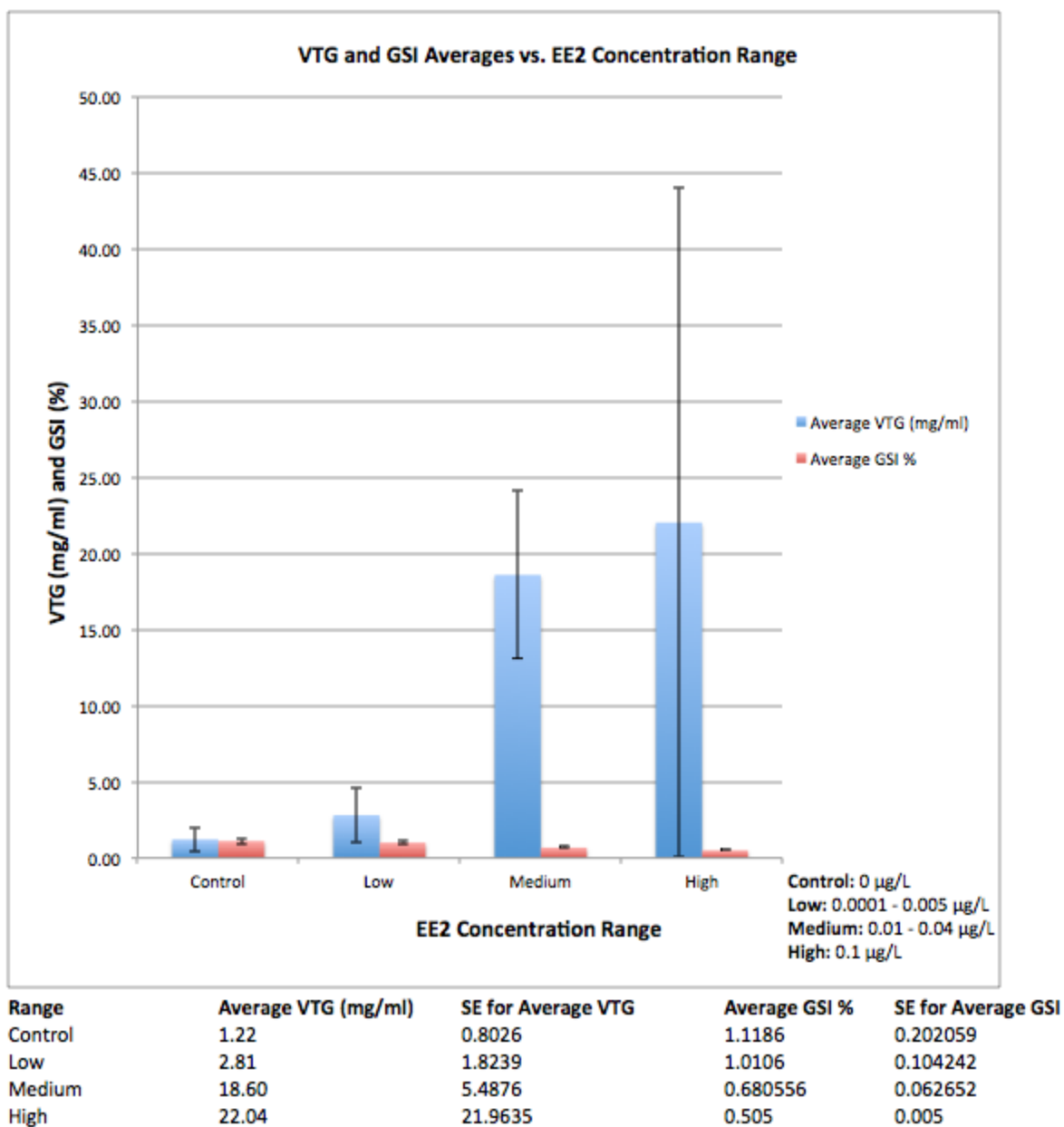


Figure 4 displays VTG concentrations and GSI values in fish exposed to the concentrations ranges of EE₂.

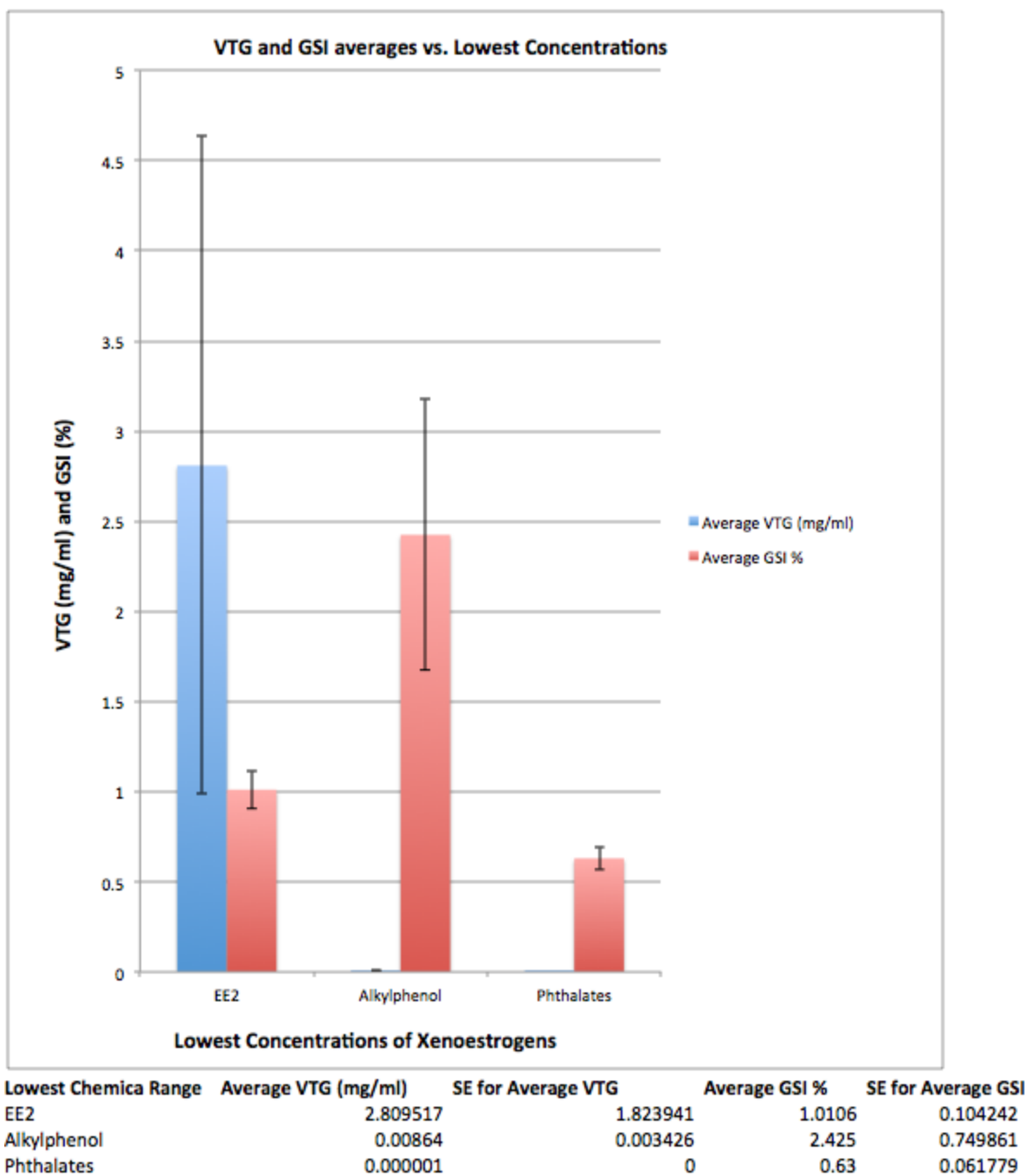


Figure 5 shows the different averages for VTG and GSI compared to the low concentrations of EE₂, alkylphenols and phthalates.

Table 1 - VTG Studies for Phthalate

	Concentration $\mu\text{g/l}$	VTG concentration
Maradona	0	0.0000001
4 days	0.1953	0.0000005
Zebrafish	0.03906	0.0000008
	0.39056	0.0000012
	3.9056	0.0000033
	39.056	0.0000049
Barse	0	0.00018
28 Days	100	0.00035
Common Carp	1000	0.00041
	5000	0.00049
Harries	0	0.00002
10 days	100	0.000045
Fat Head Minnow		

Table 1. The data collected for measuring VTG concentration and phthalate concentration, studies are differentiated by color. Information includes exposure time to phthalates as well as fish species used.

Table 2 - GSI Studies for Phthalate

Author	Concentration $\mu\text{g/l}$	GSI
Chikae 2004	0	0.97
12-15 Days	0.01	0.47
Japanese Medaka	0.1	0.6
	1	0.64
	10	0.6
Chikae 2003	0	0.85
21 Days	0.01	0.75
Japanese Medaka	0.1	0.7
	1	0.7
	10	0.78
Harries	100	1.7
10 Days		
Fathead Minnow		

Table 2. The data collected for GSI values from various studies are differentiated by different colored font. Exposure period and fish species were also included in the table.

Table 3 - VTG Data for Alkylphenol Studies

Chemical	Dose µg/L	VTG mg/ml	Study	Chemical	Dose µg/L	VTG mg/ml	Study
NP	22.035	0.00041	Soverchia <i>et al.</i> 2005	OP	0	0	Rasmussen <i>et al.</i> 2004
NP	22.035	0.00058		OP	100	17	
NP	10	0.003	Villeneuve <i>et al.</i> 2002	NP	0	0	Ruggeri <i>et al.</i> 2008
NP	3	0.01		NP	100	0.00052	
NP	1	0.005		NP	0	0.000009	Seki <i>et al.</i> 2003
NP	0.3	0.005		NP	0	0.000002	
NP	0.1	0.01		NP	3.3	0.0000008	
NP	3	0.01	Zha <i>et al.</i> 2007	NP	6.08	0.00001	
NP	10	0.04		NP	11.6	0.00003	
NP	30	0.05		NP	23.5	0.00009	
NP	0	0.009	Ishibashi <i>et al.</i> 2004	NP	44.7	0.0001	
NP	0.1	0.005		OP	0	0.000003	
NP	1	0.001		OP	0	0.0000009	
NP	10	0.0005		OP	6.94	0.000007	
NP	100	5		OP	11.4	0.00002	
NP	100	3	Harries <i>et al.</i> 2000	OP	23.7	0.00003	
NP	0	0.00003		OP	48.1	0.0002	
NP	0	0.00004		OP	94	0.0007	
NP	1	0.00008					
NP	10	0.04					
NP	100	4					

Table 3. The combined VTG data from eight alkylphenol studies, separated by author. There was a strongly positive correlation between VTG uptake and alkylphenol (NP or OP) dose.

Table 4 - GSI Data for Alkylphenol Studies

Chemical	Dose µg/L	GSI %	Study
NP	3	3.3	Zha <i>et al.</i> 2007
NP	10	4	
NP	30	4.9	Harries <i>et al.</i> 2000
NP	100	0.5	
NP	0	1	
NP	0	1.6	
NP	1	0.7	
NP	10	1.7	
NP	100	0.4	
OP	0	1.35	Rasmussen <i>et al.</i> 2004
OP	100	1.1	
OP	0	1.6	

Table 4. The combined GSI data from three alkylphenol studies, separated by author. There was a weakly negative correlation between GSI % and alkylphenol (NP or OP) dose.

Table 5 - Experimental Data for EE₂

	EE2 concentration [µg/L]	VTG (mg/ml)	GSI%
Van Den Belt, K. 2002	0	2.08	0.92
Zebrafish 6-8 mo. Old, 12 days exposure	0.01	22.5	0.58
	0.025	30	0.58
Van den Belt, K. 2001	0	0	0.833
Zebrafish 6-8 mo. Old, 21 days exposure	0.005	9	0.833
	0.01	34	0.625
	0.025	37.5	0.5
Versonnen, B.J. 2004	0	4	0.74
Zebrafish 8 mo. Old, 14 days exposure	0.001	5	0.74
	0.01	36	0.74
	0.1	44	0.51
Pawlowski, S. 2004	0	0.00028	1.25
Fathead Minnow 6-11 mo. Old, 21 days exp.	0.0001	0.0000865	1.3
	0.001	0.0015	1
	0.003	0.046	1.18
	0.01	0.0685	0.8
	0.1	0.073	0.5
Salierno, J.D. 2009	0	0.02	1.85
Fathead Minnow. 8-10 mo. Old, 21 days exposure	0.01	2.45	1.1
	0.02	2.4	0.5
	0.04	2.5	0.7

Table 5. This shows the studies used when collecting data for EE₂ and their effects on levels of VTG and GSI. Studies are provided with author, age and species of fish and duration of exposure.

Table 6 - EE₂ Control and Three Concentration Ranges

Control			
EE2 concentration [µg/L]	VTG (mg/ml)	EE2 concentration [µg/L]	GSI%
0	2.08	0	0.92
0	0.00028	0	1.25
0	0.02	0	1.85
0	4	0	0.74
0	0	0	0.833
LOW			
EE2 concentration [µg/L]	VTG (mg/ml)	EE2 concentration [µg/L]	GSI%
0.0001	0.0000865	0.0001	1.3
0.001	5	0.001	0.74
0.001	0.0015	0.001	1
0.003	0.046	0.003	1.18
0.005	9	0.005	0.833
MEDIUM			
EE2 concentration [µg/L]	VTG (mg/ml)	EE2 concentration [µg/L]	GSI%
0.01	22.5	0.01	0.58
0.01	34	0.01	0.625
0.01	36	0.01	0.74
0.01	0.0685	0.01	0.8
0.01	2.45	0.01	1.1
0.02	2.4	0.02	0.5
0.025	30	0.025	0.58
0.025	37.5	0.025	0.5
0.04	2.5	0.04	0.7
HIGH			
EE2 concentration [µg/L]	VTG (mg/ml)	EE2 concentration [µg/L]	GSI%
0.1	44	0.1	0.51
0.1	0.073	0.1	0.5

Table 6. This table shows the three concentration ranges for EE₂. The color coded data corresponds to Table 5, which shows which experiments the values came from. The amount of data for each range varied, potentially serving as a source of error in the results.

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